ASSIGNMENT No: 6

NAME: SUHANI GARG
PRN: 25070122194
BATCH: CSE B3
A.Y: 2026-27

TITLE: Write a Java program to demonstrate the use of an inner class, an anonymous class.

Problem Statement

Write a Java program to demonstrate the use of an inner class, an anonymous class.

Learning Objectives

To implement inner classes and anonymous classes for improving encapsulation and event handling in Java.

Theory

An inner class is a class defined inside another class and has access to all members of the enclosing class, including private members. Inner classes help improve code organization and logical grouping of related classes. Anonymous classes are unnamed inner classes that are instantiated only once and are mainly used for implementing interfaces or abstract classes. They simplify code by eliminating the need for separate class definitions. These concepts are widely used in Java GUI programming and event handling

Exercise

1.Create a **Vehicle program** where an inner class displays vehicle details and anonymous
class performs an action.

2. Develop a **Food Delivery Application** where an inner class handles order details andanonymous classes handle delivery status updates.

1. Create a Vehicle program where an inner class displays vehicle details and anonymous class performs an action.

CODE:

```java
class Vehicle {

    String vehicleName = "Honda City";
    String vehicleType = "Car";

    // Inner Class
    class VehicleDetails {

        void display() {
            System.out.println("Vehicle Name : " + vehicleName);
            System.out.println("Vehicle Type : " + vehicleType);
        }
    }
}

// Interface
interface Action {
    void perform();
}

public class InnerAnonymousDemo_Vehicle {

    Run | Debug
    public static void main(String args[]) {

        // Creating object of outer class
        Vehicle v = new Vehicle();

        // Creating object of inner class
        Vehicle.VehicleDetails details = v.new VehicleDetails();
        details.display();

        // Anonymous class
        Action a = new Action() {

            public void perform() {
                System.out.println(x: "Vehicle is moving...");
            }
        };

        a.perform();
    }
}
```

OUTPUT:

```
suhanigarg670@Suhanis-MacBook-Air ASSIGNMENT 6 % cd "/Users/suhanigarg670/JAVA LAB/ASSIGNMENT 6/" && javac InnerAnonymousDemo_Vehicle.java && java InnerAnonymousDemo_Vehicle
Vehicle Name : Fortuner
Vehicle Type : Car
Vehicle is moving...
suhanigarg670@Suhanis-MacBook-Air ASSIGNMENT 6 %
```

2. Develop a Food Delivery Application where an inner class handles order details and anonymous classes handle delivery status updates.

CODE:

```java
class FoodDelivery {

    String customerName = "Suhani Garg";
    String foodItem = "French Fries";

    // Inner Class
    class OrderDetails {

        void display() {
            System.out.println("Customer Name : " + customerName);
            System.out.println("Food Item : " + foodItem);
        }
    }
}

// Interface
interface DeliveryStatus {
    void updateStatus();
}

public class InnerAnonymousDemo_Food {

    Run | Debug
    public static void main(String args[]) {

        // Creating object of outer class
        FoodDelivery f = new FoodDelivery();

        // Creating object of inner class
        FoodDelivery.OrderDetails order = f.new OrderDetails();
        order.display();

        // Anonymous class
        DeliveryStatus d = new DeliveryStatus() {

            public void updateStatus() {
                System.out.println(x: "Order has been delivered.");
            }
        };

        d.updateStatus();
    }
}
```

OUTPUT:

```
<uhanigarg670/JAVA LAB/ASSIGNMENT 6/" && javac InnerAnonymousDemo
_Food.java && java InnerAnonymousDemo_Food
Customer Name : Suhani Garg
Food Item : French Fries
Order has been delivered.
suhanigarg670@Suhanis-MacBook-Air ASSIGNMENT 6 %
```

3. Practice question done in class:-

CODE:

```java
class College {

    String collegeName = "SIT Pune";

    // Inner Class
    class StudentMarks {

        void display() {
            System.out.println("Student belongs to: " + collegeName);
        }
    }
}

// Interface
interface Greeting {

    void sayHello();
}

// Main Class
public class InnerAnonymousDemo {

    public static void main(String args[]) {

        // Inner class object
        College college = new College();
        College.StudentMarks student = college.new StudentMarks();
        student.display();

        // Anonymous class
        Greeting g = new Greeting() {

            public void sayHello() {
                System.out.println("Hello from Anonymous class");
            }
        };

        g.sayHello();
    }
}
```

OUTPUT:

```
Student belongs to: SIT Pune
Hello from Anonymous class
suhanigarg670@Suhanis-MacBook-Air ASSIGNMENT 6 %
```

Conclusion:-

This assignment demonstrated the concepts of inheritance and interfaces in Java using real-world examples. Inheritance helped reuse common properties and methods, while interfaces provided a common contract for implementing di erent functionalities. These concepts improve code reusability, abstraction, and maintainability, making Java programs more organized and efficient.